

Supplementary Material

Contrast-Level Aggregate-Data Dose–Response Meta-Analysis for Heterogeneous Exposure Intervals

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Supplementary Methods S1: Regional distance-data construction

To estimate the parameters α and θ of the Gamma distance distribution, regional residential and animal-facility or agricultural-site location data were collected from publicly available data. The eight studies included in the meta-analysis covered four primary geographic regions: North Carolina, Wisconsin, Pennsylvania, and the Netherlands. Among the Netherlands studies, most were concentrated in Noord-Brabant and Limburg.

Publicly available spatial data for the U.S. regions were obtained from ArcGIS, including residential census-block data and facility or site records. Residential information was drawn from U.S. Census data, including population and housing counts and geographic coordinates [1]. The facility classification schemes differed across state datasets and referred primarily to animal species or production category. Categories most directly related to animal-production exposures were retained; dairy- and egg-specific categories were excluded when identifiable. North Carolina and Wisconsin used permitted-facility layers [2, 3]. Pennsylvania used the public CRAFT agricultural-producer directory, separated into dairy, egg, and meat categories; it is not the Pennsylvania DEP CAFO permit layer and is therefore a geospatial proxy [4, 5]. Spatial processing used ArcGIS Pro [6].

For the Netherlands, corresponding data covering all study regions were not publicly available. Residential data came from the CBS Wijk- en Buurtkaart dataset, and the site layer describes agricultural businesses rather than verified AFO permits [7, 8]. Records were filtered by Dutch livestock-related keywords in the facility-type and category fields. The available municipality of Eijsden-Margraten was therefore used to estimate a proxy Netherlands distance distribution. Its fitted mean of approximately 0.58 miles should not be interpreted as national residential distance to a verified AFO.

Within each region, distances between residential units and recorded facilities or proxy sites were calculated from latitude and longitude using *distHaversine()* in the **geosphere** R package [9]. The minimum distance was retained and the regional Gamma parameters were estimated by population-weighted maximum likelihood.

The principal pooled specification equally weighted the four parameter sets from the Netherlands, North Carolina, Pennsylvania, and Wisconsin, because these are the regions represented by included studies. Its shape was their arithmetic mean; its mean distance was the arithmetic mean of $\alpha_r\theta_r$; and its scale was the pooled mean divided by the pooled shape. Iowa served only as an additional public U.S. reference region in a separate five-region external scenario [10, 11]. A source-quality sensitivity retained the North Carolina and Wisconsin permitted-facility parameters, pooled those two Gamma fits, and used that pooled working distribution in place of the Pennsylvania and Netherlands proxy-layer parameters. This scenario assesses dependence on first-stage source type; it does not assume that the two U.S. registries are geographically transportable.

The regional fitting scripts requested numerical Hessians, but neither the Hessian matrices nor the observation-level distance files were retained in the reproducibility bundle. First-stage covariance propagation therefore could not be reconstructed from the stored shape and scale point estimates. The reported application uncertainty is conditional on those parameters, and scenario analyses address broader source and proxy uncertainty.

The analysis workbook records two mapping types. Type 1 retains the reported comparison and reference as an explicit pair of positive- or zero-width supports. Type 2 is a proxy conversion for a binary indicator defined by a radius r : the analysis support is $(0, r]$ and the complementary support is $(r, d_{\max}]$. Here $d_{\max}$ is the finite operational bound recorded in the co-author-curated extraction; it may be a graph-axis, geographic, or other analyst-imposed study-population bound and is not represented as a reported participant maximum unless the source establishes that fact. The primary AFO working analysis retains all 33 mapped contrasts. The OR-only restriction removes the four animal-count hazard-ratio proxies, and the strict explicit- support OR restriction additionally removes Type 2 mappings. Explicit species- specific distance intervals remain Type 1; species-specific binary-buffer or presence indicators remain Type 2. For source categories genuinely reported as right-open, the strict-restriction sensitivity replaced $(r, d_{\max}]$ with (r, ∞) ; the Gamma tail mean is finite and the rounded result was unchanged.

Supplementary Methods S2: External residential-radon distributions

The independent illustration used approximately contemporaneous national residential-radon distributions for the primary analysis and current UNSCEAR parameters for sensitivity. The disease-effect table supplied adjusted ORs, confidence intervals, and support bounds only; none of the survey parameters were estimated from those outcomes or from exposure-category frequencies. Integer category labels were interpreted as rounded measurements and converted to contiguous half-unit boundaries (for example, 74–147 became $(73.5, 147.5]$); explicitly reported decimal boundaries were retained.

Table S1: Country-specific Lognormal working distributions used in the residential-radon illustration.

Country	Historical GM	Historical GSD	UNSCEAR 2024 GM	UNSCEAR 2024 GSD
China	20	2.2	40	2.4
United States	25	3.1	25	3.1
Sweden	56	3.146	63	2.3
Finland	84	2.1	62	2.5
United Kingdom	14	3.2	15	2.2
Japan	13	1.8	11	2.1
Germany	37	2.0	41	2.0
Italy	52	2.1	50	2.2
Spain	46	2.9	25	3.4

GM is geometric mean in Bq/m³ and GSD is geometric standard deviation. Historical values are from UNSCEAR 2000 or the WHO handbook [12, 13]; 2024 values are from UNSCEAR Table 2 [14]. Sweden’s historical GSD was derived under Lognormality from the reported arithmetic mean 108 and geometric mean 56 as $\exp\{2\log(108/56)\}^{1/2} = 3.14591897$ (3.146 to three decimals).

Supplementary Methods S3: Targeted method-positioning search

To assess the methodological positioning of the proposed framework, we conducted a targeted, non-systematic search on August 14, 2026. The PubMed queries were: (1) ("dose-response"[Title/Abstract] OR "dose response"[Title/Abstract]) AND (meta-analysis[Title/Abstract] OR meta-regression[Title/Abstract]) AND (grouped[Title/Abstract] OR categorical[Title/Abstract] OR interval[Title/Abstract]); (2) ("dose-response meta-analysis"[Title/Abstract]) AND (midpoint[Title/Abstract] OR "assigned dose"[Title/Abstract] OR "conditional mean"[Title/Abstract]); and (3) (meta-analysis[Title/Abstract] AND ("multiple cutpoints"[Title/Abstract] OR "different cutpoints"[Title/Abstract] OR "heterogeneous cutpoints"[Title/Abstract])). Publisher-indexed and citation searches used the exact strings "dose response meta-analysis" "adjusted odds ratio" interval midpoint, "grouped exposure" "conditional mean" meta-analysis, "interval-censored" exposure meta-analysis, and "multiple cutpoints" meta-analysis. We supplemented these searches by backward and forward citation searching from conventional summarized-data DRMA papers and the closest grouped-exposure, interval-censoring, and multiple-cutpoint methods [15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25].

We retained statistical methods addressing synthesis or dose assignment for grouped continuous exposures and used their reported data requirements to construct Supplementary Table S2. This search was intended to position the method rather than to estimate the prevalence of methods, and it should not be interpreted as a systematic review. Among the methods identified, none accepted the complete input combination considered here: already-adjusted arbitrary-pair contrasts, observation-specific comparison and reference supports, mixtures of positive- and zero-width supports, within-study dependence, and no exposure-group frequencies or exposure–outcome counts. The manuscript therefore uses the qualified wording “no prior method we identified.”

Table S2: Data and structural requirements of close dose–response approaches.

Approach	Required reported information	Exposure/reference structure	Relation to the present setting
Point-dose DRMA [15, 16]	Effect estimates, assigned doses, and covariance	Point supports, usually relative to a study reference	Exact all-zero-width special case under the same basis and covariance
Shi–Copas trend [21]	Category-level risk information	Aggregated levels through a reference structure	Not formulated for arbitrary adjusted pairs with mixed supports
Different/multiple-cutpoint methods [22, 23]	Relative risks or prognostic effects and covariance	Cutpoint-specific effects, ordinarily referent-based	Model cutpoint heterogeneity but do not map arbitrary support pairs
Conventional one-/two-stage DRMA [24]	Dose-specific estimates with assigned category values	Categories relative to a study reference	Operates after the category-dose assignment at issue here
Categorical-to-continuous conversion [25]	Categorical effects, intervals, and event counts	Midpoints; category-specific per-unit coefficients	Uses event-count weights and independence rather than arbitrary-pair covariance
Grouped-exposure likelihood [17]	Joint grouped exposure and disease information	Likelihood from grouped exposure–outcome data	Cannot be reconstructed from adjusted contrasts and confidence intervals alone
Interval-mean pre-analysis [18, 19]	Bounds and exposure-group counts or proportions	Group frequencies estimate interval means	Can supply support means when frequencies exist; unavailable here
Grouped-dose assignment [20]	Intervals and summarized category responses	Representative values before regression	Requires response summaries unavailable from adjusted pairwise contrasts alone
Proposed operator	Adjusted contrast, uncertainty, support pair, and working exposure distribution	Each row may compare its own positive- or zero-width supports	Retains covariance without individual data, group counts, or a common reference

Supplementary Methods S4: Proofs moved from the main text

Proposition 1. Linearity of conditional expectation gives $E\{\gamma_k + \beta_k^\top \mathbf{h}(D) \mid D \in S\} = \gamma_k + \beta_k^\top E\{\mathbf{h}(D) \mid D \in S\}$. For a zero-width support, continuity and the shrinking-interval limit give $\mathbf{m}_k\{\mathcal{I}(d, d)\} = \mathbf{h}(d)$, so every downstream point-dose DRMA expression is unchanged. Under a Uniform conditional distribution and a linear basis, direct integration gives $\int_a^b u(b-a)^{-1} du = (a+b)/2$; hence every design contrast and downstream midpoint-DRMA expression is unchanged. Because the mapping is defined support by support, points and intervals may occur in any mixture. $\square$

Conditional-mean projection and aggregation coherence. For any support S with positive probability and finite conditional second moment,

$$E\{(D - c)^2 \mid D \in S\} = \text{Var}(D \mid D \in S) + \{c - E(D \mid D \in S)\}^2.$$

Thus the conditional mean is the unique minimum-MSE constant representation under squared-error loss; midpoint or endpoint scoring incurs the squared difference from that mean. If S is partitioned into disjoint $S_1, \dots, S_R$, the law of total expectation gives

$$E(D \mid D \in S) = \sum_{r=1}^R P(D \in S_r \mid D \in S) E(D \mid D \in S_r).$$

These are representation properties, not minimum-MSE claims for the fitted slope under arbitrary outcome links. $\square$

Proposition 2. Positive definiteness of $\mathbf{V}_k$ makes every $\mathbf{x}_k^\top \mathbf{V}_k^{-1} \mathbf{x}_k$ nonnegative and zero only when $\mathbf{x}_k = \mathbf{0}$. Hence $\mathcal{J}(\tau) > 0$ if and only if at least one design contrast is nonzero. For specified τ , substitution of $E(\mathbf{y}_k) = \beta_1 \mathbf{x}_k$ in the GLS numerator gives $E\{\hat{\beta}_1(\tau)\} = \beta_1$, and independence of the study blocks gives

$$\text{Var}\{\hat{\beta}_1(\tau)\} = \mathcal{J}(\tau)^{-2} \sum_k \mathbf{x}_k^\top \mathbf{V}_k^{-1} \mathbf{V}_k \mathbf{V}_k^{-1} \mathbf{x}_k = \mathcal{J}(\tau)^{-1}.$$

This calculation is conditional on specified τ and does not assert exact finite-sample unbiasedness after substituting $\hat{\tau}$.

An affine shift cancels in support differences. Under $D^* = a + cD$, $c > 0$, $\mathbf{x}_k^* = c\mathbf{x}_k$, $\beta_1^* = \beta_1/c$, and $\tau^* = \tau/c$. Therefore, at corresponding parameter values,

$$\mathbf{V}_k^*(\tau/c) = \boldsymbol{\Sigma}_k + (\tau/c)^2 (c\mathbf{x}_k)(c\mathbf{x}_k)^\top = \mathbf{V}_k(\tau), \quad \mathcal{J}^*(\tau/c) = c^2 \mathcal{J}(\tau).$$

It follows directly that $\hat{\beta}_1^*(\tau/c) = \hat{\beta}_1(\tau)/c$ and that the fitted residual vectors, quadratic forms, and log determinants in the restricted likelihood are unchanged. The final information term changes by $-\frac{1}{2} \log\{c^2 \mathcal{J}(\tau)\} + \frac{1}{2} \log\{\mathcal{J}(\tau)\} = -\log c$. Thus $\ell_R^*(\tau/c) = \ell_R(\tau) - \log c$. The added constant does not affect the profile optimizer or likelihood differences, so $\hat{\tau}^* = \hat{\tau}/c$, $\hat{\beta}_1^* = \hat{\beta}_1/c$, and profile-likelihood limits transform by the same factor. $\square$

Proposition 4. Conditional on $\boldsymbol{\psi}$, the meta-analytic estimator contributes $\hat{\mathbf{C}}_{\text{cond}}$. A first-order Taylor expansion of $\hat{\boldsymbol{\vartheta}}(\hat{\boldsymbol{\psi}})$ around the true external parameter gives $\hat{\mathbf{G}} \hat{\mathbf{C}}_{\boldsymbol{\psi}} \hat{\mathbf{G}}^\top$; independence removes the cross-covariance term. $\square$

Supplementary Methods S5: Numerical implementation and validation

The production estimator is implemented in `interval.drma.core.py`; application and simulation programs import that core module. Design values are divided internally by their root-mean-square value, REML is optimized on the dimensionless scale, and slopes, heterogeneity, standard errors, and profile limits are transformed back. The exact $\tau = 0$ criterion is compared with the interior optimum; the upper bracket expands until the objective is increasing; optimizer status, finite criteria, boundary solutions, and the number of expansions are stored in the JSON output. Gamma and Lognormal conditional means use stable tail probabilities and raise an error if an interval probability cannot be resolved.

The deterministic suite contains 20 tests, including exposure multipliers from 10^{-6} through 10^6 , exact $\tau = 0$, near-zero and large heterogeneity, row- and study-order invariance, point/midpoint special cases, and extreme open-tail or narrow supports. All 20 tests passed. The archived analyses were run on Windows 11 with Python 3.13.9, NumPy 2.3.4, SciPy 1.16.3, openpyxl 3.1.5, and Matplotlib 3.10.7. Application bootstrap seeds were 314159 (AFO) and 161803 (radon); the ten-study template seed was 8675309. All remaining deterministic cell seeds are recorded directly in `mixed.support.simulation.py`.

Supplementary Tables

Table S3: Leave-one-study-out influence analysis for the AFO illustration.

Omitted study	$\hat{\beta}_1$	95% CI	$\hat{\tau}$
Rasmussen et al.	-0.071	(-0.192, 0.051)	0.147
Smit et al. (2014)	-0.096	(-0.191, -0.001)	0.112
Smit et al. (2012)	-0.077	(-0.182, 0.029)	0.133
Freidl et al.	-0.045	(-0.143, 0.053)	0.117
Schultz et al.	-0.038	(-0.120, 0.044)	0.093
Post et al.	-0.084	(-0.197, 0.029)	0.137
Son et al.	-0.082	(-0.199, 0.036)	0.141
Simoës et al.	-0.064	(-0.185, 0.057)	0.146

Each row refits the primary random-effects model after omitting the named independent study block.

Table S4: Leave-one-study-out influence analysis for the residential-radon illustration.

Omitted study	OR per 100 Bq/m ³	95% CI
Blot et al.	1.093	(1.046, 1.141)
Schoenberg et al.	1.079	(1.030, 1.131)
Pershagen et al. (1992)	1.073	(1.022, 1.126)
Alavanja et al. (1994)	1.077	(1.025, 1.132)
Pershagen et al. (1994)	1.077	(1.018, 1.139)
Auvinen et al.	1.091	(1.039, 1.146)
Ruosteenoja et al.	1.080	(1.029, 1.133)
Darby et al.	1.094	(1.047, 1.144)
Alavanja et al. (1999)	1.087	(1.037, 1.139)
Field et al.	1.077	(1.024, 1.132)
Sobue et al.	1.082	(1.033, 1.134)
Kreienbrock et al.	1.083	(1.032, 1.136)
Lagarde et al.	1.075	(1.024, 1.128)
Pisa et al.	1.083	(1.033, 1.136)
Wang et al.	1.072	(1.020, 1.127)
Barros-Dios et al.	1.077	(1.027, 1.129)

Values are random-effects estimates under the historical country-specific Lognormal distributions and $\rho = 0.5$.

Table S5: Random-effects sensitivity analysis under alternative working correlations and Gamma distance distributions.

Distribution specification	ρ	$\hat{\beta}_1$	95% CI	$\hat{\tau}$	95% CI
Regional	0	-0.058	(-0.156, 0.039)	0.129	(0.073, 0.261)
Regional	0.25	-0.063	(-0.162, 0.036)	0.127	(0.065, 0.268)
Regional	0.5	-0.069	(-0.170, 0.031)	0.129	(0.063, 0.274)
Regional	0.75	-0.078	(-0.183, 0.027)	0.138	(0.072, 0.285)
Regional	0.9	-0.084	(-0.193, 0.025)	0.146	(0.084, 0.292)
Four-region pooled	0	-0.042	(-0.103, 0.019)	0.086	(0.049, 0.171)
Four-region pooled	0.25	-0.039	(-0.094, 0.016)	0.077	(0.035, 0.162)
Four-region pooled	0.5	-0.038	(-0.090, 0.014)	0.072	(0.026, 0.158)
Four-region pooled	0.75	-0.042	(-0.098, 0.014)	0.078	(0.036, 0.165)
Four-region pooled	0.9	-0.049	(-0.112, 0.015)	0.090	(0.052, 0.179)
Five-region + Iowa	0	-0.043	(-0.104, 0.019)	0.087	(0.049, 0.172)
Five-region + Iowa	0.25	-0.040	(-0.096, 0.016)	0.077	(0.036, 0.163)
Five-region + Iowa	0.5	-0.039	(-0.092, 0.014)	0.073	(0.028, 0.159)
Five-region + Iowa	0.75	-0.043	(-0.100, 0.014)	0.079	(0.037, 0.166)
Five-region + Iowa	0.9	-0.050	(-0.114, 0.014)	0.091	(0.052, 0.180)
Four-region scale -20%	0	-0.043	(-0.105, 0.018)	0.087	(0.049, 0.173)
Four-region scale -20%	0.25	-0.041	(-0.098, 0.016)	0.079	(0.037, 0.165)
Four-region scale -20%	0.5	-0.040	(-0.094, 0.014)	0.074	(0.031, 0.161)
Four-region scale -20%	0.75	-0.045	(-0.103, 0.013)	0.080	(0.039, 0.168)
Four-region scale -20%	0.9	-0.051	(-0.116, 0.013)	0.092	(0.053, 0.181)
Four-region scale +20%	0	-0.041	(-0.102, 0.020)	0.086	(0.048, 0.171)
Four-region scale +20%	0.25	-0.038	(-0.092, 0.017)	0.076	(0.033, 0.161)
Four-region scale +20%	0.5	-0.036	(-0.087, 0.014)	0.070	(0.022, 0.156)
Four-region scale +20%	0.75	-0.041	(-0.096, 0.014)	0.077	(0.034, 0.164)
Four-region scale +20%	0.9	-0.047	(-0.110, 0.016)	0.090	(0.051, 0.178)
NC/WI registry proxy for PA/NL	0	-0.040	(-0.100, 0.021)	0.086	(0.048, 0.171)
NC/WI registry proxy for PA/NL	0.25	-0.036	(-0.090, 0.018)	0.075	(0.032, 0.161)
NC/WI registry proxy for PA/NL	0.5	-0.035	(-0.085, 0.015)	0.069	(0.016, 0.155)
NC/WI registry proxy for PA/NL	0.75	-0.039	(-0.094, 0.015)	0.076	(0.033, 0.163)
NC/WI registry proxy for PA/NL	0.9	-0.046	(-0.109, 0.017)	0.089	(0.051, 0.177)

Table S6: Fixed-effect sensitivity analysis under alternative working correlations and Gamma distance distributions.

Distribution specification	ρ	$\hat{\beta}_1$	95% CI
Regional	0	-0.007	(-0.013, -0.002)
Regional	0.25	-0.009	(-0.016, -0.002)
Regional	0.5	-0.011	(-0.020, -0.003)
Regional	0.75	-0.015	(-0.024, -0.006)
Regional	0.9	-0.020	(-0.028, -0.012)
Four-region pooled	0	-0.008	(-0.012, -0.005)
Four-region pooled	0.25	-0.009	(-0.013, -0.004)
Four-region pooled	0.5	-0.010	(-0.015, -0.005)
Four-region pooled	0.75	-0.012	(-0.017, -0.006)
Four-region pooled	0.9	-0.014	(-0.019, -0.009)
Five-region + Iowa	0	-0.009	(-0.012, -0.005)
Five-region + Iowa	0.25	-0.009	(-0.014, -0.005)
Five-region + Iowa	0.5	-0.010	(-0.016, -0.005)
Five-region + Iowa	0.75	-0.012	(-0.018, -0.006)
Five-region + Iowa	0.9	-0.014	(-0.019, -0.009)
Four-region scale -20%	0	-0.009	(-0.013, -0.005)
Four-region scale -20%	0.25	-0.010	(-0.015, -0.005)
Four-region scale -20%	0.5	-0.011	(-0.017, -0.005)
Four-region scale -20%	0.75	-0.012	(-0.018, -0.006)
Four-region scale -20%	0.9	-0.014	(-0.019, -0.009)
Four-region scale +20%	0	-0.007	(-0.011, -0.004)
Four-region scale +20%	0.25	-0.008	(-0.012, -0.004)
Four-region scale +20%	0.5	-0.009	(-0.014, -0.005)
Four-region scale +20%	0.75	-0.011	(-0.016, -0.006)
Four-region scale +20%	0.9	-0.014	(-0.018, -0.009)
NC/WI registry proxy for PA/NL	0	-0.007	(-0.010, -0.004)
NC/WI registry proxy for PA/NL	0.25	-0.008	(-0.011, -0.004)
NC/WI registry proxy for PA/NL	0.5	-0.009	(-0.013, -0.004)
NC/WI registry proxy for PA/NL	0.75	-0.010	(-0.015, -0.006)
NC/WI registry proxy for PA/NL	0.9	-0.013	(-0.017, -0.009)

Table S7: Bias (RMSE) in the Gamma-true mixed-support simulation. Each setting used $R = 5,000$ replicates.

$\beta_{1,\text{true}}$	τ_{true}	Conditional mean		Midpoint	
		$\hat{\beta}_1$	$\hat{\tau}$	$\hat{\beta}_1$	$\hat{\tau}$
−0.035	0.065	0.001 (0.027)	−0.005 (0.027)	0.028 (0.028)	−0.050 (0.050)
−0.069	0.065	−0.001 (0.027)	−0.006 (0.028)	0.055 (0.055)	−0.047 (0.048)
−0.139	0.065	< 0.001 (0.028)	−0.005 (0.027)	0.108 (0.109)	−0.039 (0.041)
−0.035	0.129	0.001 (0.047)	−0.006 (0.039)	0.027 (0.030)	−0.096 (0.098)
−0.069	0.129	−0.001 (0.046)	−0.006 (0.040)	0.053 (0.054)	−0.094 (0.096)
−0.139	0.129	< 0.001 (0.046)	−0.007 (0.039)	0.106 (0.107)	−0.087 (0.090)
−0.035	0.259	< 0.001 (0.084)	−0.010 (0.065)	0.026 (0.038)	−0.177 (0.182)
−0.069	0.259	0.001 (0.085)	−0.008 (0.066)	0.052 (0.059)	−0.177 (0.182)
−0.139	0.259	0.001 (0.085)	−0.009 (0.066)	0.102 (0.106)	−0.172 (0.178)

Table S8: Bias (RMSE) in the Uniform-true mixed-support simulation. Each setting used $R = 5,000$ replicates.

$\beta_{1,\text{true}}$	τ_{true}	Conditional mean		Midpoint	
		$\hat{\beta}_1$	$\hat{\tau}$	$\hat{\beta}_1$	$\hat{\tau}$
−0.035	0.065	< 0.001 (0.022)	−0.002 (0.017)	< 0.001 (0.022)	−0.002 (0.017)
−0.069	0.065	−0.001 (0.022)	−0.002 (0.017)	−0.001 (0.022)	−0.002 (0.017)
−0.139	0.065	> −0.001 (0.022)	−0.002 (0.017)	> −0.001 (0.022)	−0.002 (0.017)
−0.035	0.129	−0.001 (0.042)	−0.004 (0.031)	−0.001 (0.042)	−0.004 (0.031)
−0.069	0.129	0.001 (0.041)	−0.004 (0.031)	0.001 (0.041)	−0.004 (0.031)
−0.139	0.129	< 0.001 (0.041)	−0.004 (0.031)	< 0.001 (0.041)	−0.004 (0.031)
−0.035	0.259	−0.001 (0.082)	−0.008 (0.062)	−0.001 (0.082)	−0.008 (0.062)
−0.069	0.259	−0.001 (0.083)	−0.007 (0.061)	−0.001 (0.083)	−0.007 (0.061)
−0.139	0.259	< 0.001 (0.082)	−0.005 (0.060)	< 0.001 (0.082)	−0.005 (0.060)

Table S9: Point and interval performance in mixed-support stress tests. Unless stated otherwise, $\beta_{1,\text{true}} = -0.069$, $\tau_{\text{true}} = 0.129$, and $R = 5,000$.

Scenario	Bias (RMSE), $\hat{\beta}_1$	Bias (RMSE), $\hat{\tau}$	Normal cov.	t cov.	$\tau = 0$ rate	Conv.
Correct Gamma working model	0.001 (0.046)	-0.007 (0.039)	0.907	0.938	0.002	1.000
Lognormal true; Gamma fitted	-0.001 (0.046)	-0.004 (0.040)	0.914	0.945	0.003	1.000
Uniform true; Gamma fitted	-0.290 (0.399)	0.728 (0.800)	0.823	0.904	0.000	1.000
True $d_{\max}$ 20% lower	< 0.001 (0.046)	-0.007 (0.040)	0.910	0.939	0.001	1.000
True $d_{\max}$ 20% higher	> -0.001 (0.046)	-0.007 (0.040)	0.908	0.941	0.003	1.000
True $\rho = 0.3$; fitted $\rho = 0.5$	-0.001 (0.046)	-0.006 (0.040)	0.908	0.941	0.003	1.000
True $\rho = 0.7$; fitted $\rho = 0.5$	< 0.001 (0.046)	-0.006 (0.039)	0.912	0.946	0.002	1.000
Individual-level logistic aggregation; $0.5 \times \beta_1$	0.003 (0.045)	-0.009 (0.040)	0.909	0.942	0.003	1.000
Individual-level logistic aggregation; $1.0 \times \beta_1$	0.004 (0.045)	-0.010 (0.040)	0.911	0.941	0.003	1.000
Individual-level logistic aggregation; $2.0 \times \beta_1$	0.006 (0.044)	-0.012 (0.041)	0.909	0.939	0.003	1.000
Gamma shape/scale uncertainty; CV 10%	-0.001 (0.046)	-0.005 (0.039)	0.911	0.940	0.002	1.000
Gamma shape/scale uncertainty; CV 20%	0.001 (0.046)	-0.008 (0.040)	0.903	0.940	0.003	1.000
Nonlinear individual-level logit; $c = 0.05$	0.024 (0.052)	-0.011 (0.040)	0.868	0.905	0.002	1.000
Nonlinear individual-level logit; $c = 0.10$	0.044 (0.063)	-0.008 (0.038)	0.807	0.861	0.003	1.000

Monte Carlo standard errors for coverage were at most 0.0056. The t interval uses $K - 1 = 9$ degrees of freedom.

Table S10: Interval performance in the 3×3 random-effects simulation grid ($K = 10$, $R = 5,000$).

$\beta_{1,\text{true}}$	τ_{true}	Normal cov.	t cov.	Normal width	t width	$\tau = 0$ rate	Conv.
-0.035	0.065	0.8678	0.8984	0.0980	0.1131	0.0310	1.000
-0.069	0.065	0.8670	0.9012	0.0983	0.1134	0.0300	1.000
-0.139	0.065	0.8720	0.9040	0.0982	0.1133	0.0294	1.000
-0.035	0.129	0.9120	0.9414	0.1755	0.2026	0.0024	1.000
-0.069	0.129	0.9102	0.9438	0.1742	0.2010	0.0030	1.000
-0.139	0.129	0.9042	0.9390	0.1749	0.2018	0.0034	1.000
-0.035	0.259	0.9234	0.9516	0.3248	0.3748	0.0002	1.000
-0.069	0.259	0.9060	0.9456	0.3241	0.3741	0.0000	1.000
-0.139	0.259	0.9218	0.9536	0.3260	0.3763	0.0000	1.000

Coverage Monte Carlo standard errors ranged from 0.0030 to 0.0048. The t interval uses 9 degrees of freedom.

Table S11: Null-slope type-I error across heterogeneity levels ($K = 10$, $R = 5,000$).

τ_{true}	Normal type I	t type I	Normal width	t width	$\tau = 0$ rate	Conv.
0.065	0.1330	0.1026	0.0983	0.1134	0.0284	1.000
0.129	0.0992	0.0660	0.1725	0.1991	0.0030	1.000
0.259	0.0900	0.0516	0.3238	0.3737	0.0000	1.000

Type-I-error Monte Carlo standard errors were 0.0048, 0.0042, and 0.0040 for the Normal rows; the corresponding t -interval values are available in the machine-readable results.

Table S12: Study information weights and covariance conditioning at the primary working correlation.

Analysis	K	Information-weight range	Max. $\kappa(\Sigma_k)$	Max. $\kappa(V_k)$
AFO primary working analysis	8	0.047–0.157	25.97	4994.48
AFO explicit-support OR restriction	4	0.226–0.279	8.40	69.74
Residential radon	16	0.0005–0.1846	45.80	59.42

Weights are each study’s contribution to $\sum_k \mathbf{x}_k^\top V_k^{-1} \mathbf{x}_k$, normalized to sum to one; κ is the 2-norm condition number. The four largest primary AFO weights were 0.157 (Son), 0.156 (Rasmussen), 0.154 (Simoes), and 0.134 (Schultz). The large primary $\kappa(V_k)$ occurs in the Son block and is reported explicitly; the one-most-precise-contrast-per-study sensitivity avoids reliance on its within-study multicontrast geometry. AFO explicit-support restriction weights were 0.279 (Rasmussen), 0.236 (Smit 2014), 0.226 (Freidl), and 0.259 (Schultz). The four largest radon weights were 0.185 (Perschagen 1994), 0.126 (Auvinen), 0.122 (Wang), and 0.117 (Darby 1998). Across $\rho = 0$ to 0.9, the radon OR per 100 Bq/m³ changed from 1.109 to 1.035 as the covariance reweighted multicontrast blocks. The one-most-precise-contrast-per-study radon sensitivity gave OR 1.163 (study-count- t interval 1.064–1.271), compared with the CR2 sensitivity of 1.081 (1.028–1.137).

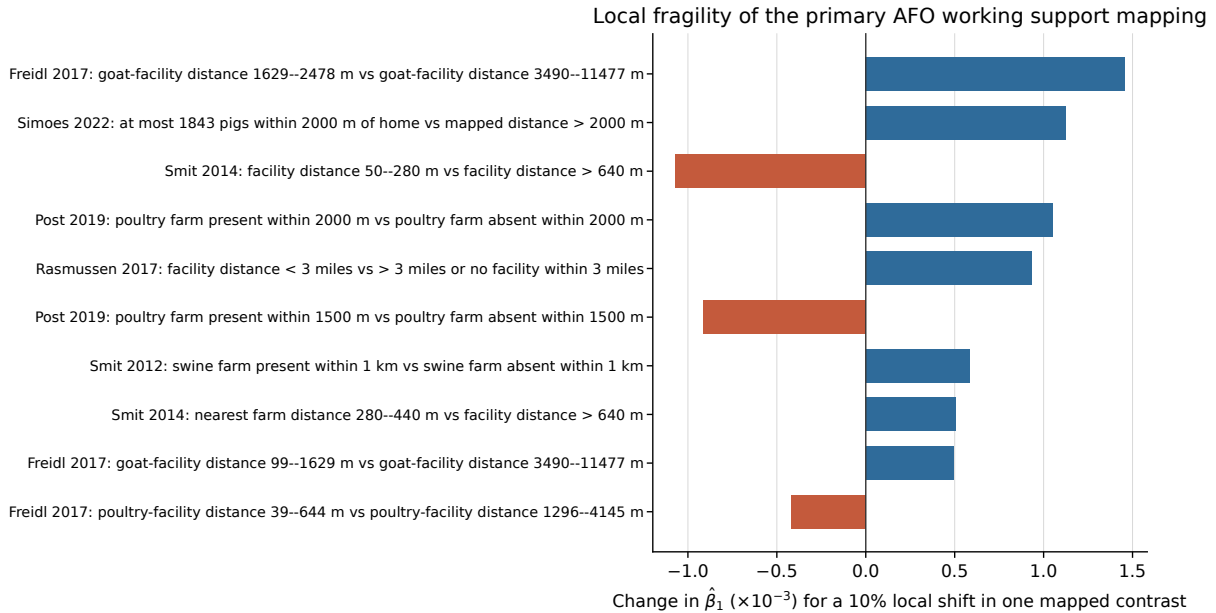


Figure S1: Local fragility of the primary eight-study AFO working support-to-design mapping. Each bar is the change in the pooled slope after a first-order 10% perturbation of one mapped interval contrast, holding all other mapped contrasts and reported outcomes fixed. The ten largest absolute changes are displayed; exact point-to-point contrasts are excluded because their design values require no interval mapping.

Table S13: Per-comparison audit of AFO support mappings under the primary regional Gamma distributions.

Study	Region	Map	Role	Comparison support	Reference support	$d_{\max}$	Source of finite maximum	$\mu(A)$	$\mu(B)$
Rasmussen	PA	T1	Explicit OR	(0.00, 3.00]	(3.00, 40.00]	40.00	Source graph label	1.77	6.04
Smit 2014	Netherlands	T2	Buffer OR	(0.00, 0.31]	(0.31, 15.00]	15.00	Same regional source	0.20	0.70
Smit 2014	Netherlands	T1	Explicit OR	(0.27, 0.40]	(0.40, 15.00]	15.00	Same study/source maximum	0.34	0.77
Smit 2014	Netherlands	T1	Explicit OR	(0.18, 0.27]	(0.40, 15.00]	15.00	Same study/source maximum	0.23	0.77
Smit 2014	Netherlands	T1	Explicit OR	(0.03, 0.18]	(0.40, 15.00]	15.00	Same study/source maximum	0.13	0.77
Smit 2012	Netherlands	T2	Buffer OR	(0.00, 0.62]	(0.62, 15.00]	15.00	Source graph label	0.36	0.95
Freidl 2017	Netherlands	T2	Buffer OR	(0.00, 0.62]	(0.62, 7.46]	7.46	Reported category maximum	0.36	0.95
Freidl 2017	Netherlands	T2	Buffer OR	(0.00, 0.93]	(0.93, 7.46]	7.46	Reported category maximum	0.47	1.23
Freidl 2017	Netherlands	T2	Buffer OR	(0.00, 0.31]	(0.31, 7.46]	7.46	Reported category maximum	0.20	0.70
Freidl 2017	Netherlands	T1	Explicit OR	(0.57, 0.81]	(0.81, 2.58]	7.46	Reported category maximum	0.68	1.11
Freidl 2017	Netherlands	T1	Explicit OR	(1.54, 2.17]	(2.17, 7.13]	7.46	Reported category maximum	1.74	2.43
Freidl 2017	Netherlands	T1	Explicit OR	(0.40, 0.57]	(0.81, 2.58]	7.46	Reported category maximum	0.48	1.11
Freidl 2017	Netherlands	T1	Explicit OR	(1.01, 1.54]	(2.17, 7.13]	7.46	Reported category maximum	1.20	2.43
Freidl 2017	Netherlands	T1	Explicit OR	(0.02, 0.40]	(0.81, 2.58]	7.46	Reported category maximum	0.26	1.11
Freidl 2017	Netherlands	T1	Explicit OR	(0.06, 1.01]	(2.17, 7.13]	7.46	Reported category maximum	0.49	2.43
Freidl 2017	Netherlands	T2	Buffer OR	(0.00, 1.24]	(1.24, 7.46]	7.46	Reported category maximum	0.53	1.52
Schultz 2019	Wisconsin	T1	Point OR	{1.00}	{5.00}	—	Not required	1.00	5.00
Schultz 2019	Wisconsin	T1	Point OR	{1.50}	{5.00}	—	Not required	1.50	5.00
Schultz 2019	Wisconsin	T1	Point OR	{2.00}	{5.00}	—	Not required	2.00	5.00
Schultz 2019	Wisconsin	T1	Point OR	{2.50}	{5.00}	—	Not required	2.50	5.00
Schultz 2019	Wisconsin	T1	Point OR	{3.00}	{5.00}	—	Not required	3.00	5.00
Post 2019	Netherlands	T2	Buffer OR	(0.00, 0.31]	(0.31, 7.46]	7.46	Source descriptive plot	0.20	0.70
Post 2019	Netherlands	T2	Buffer OR	(0.00, 0.62]	(0.62, 7.46]	7.46	Source descriptive plot	0.36	0.95
Post 2019	Netherlands	T2	Buffer OR	(0.00, 0.93]	(0.93, 7.46]	7.46	Source descriptive plot	0.47	1.23
Post 2019	Netherlands	T2	Buffer OR	(0.00, 1.24]	(1.24, 7.46]	7.46	Source descriptive plot	0.53	1.52
Son 2021	NC	T2	Buffer OR	(0.00, 3.11]	(3.11, 43.00]	43.00	Source graph label	1.89	10.76
Son 2021	NC	T2	Buffer OR	(0.00, 6.21]	(6.21, 43.00]	43.00	Source graph label	3.53	13.15
Son 2021	NC	T2	Buffer OR	(0.00, 9.32]	(9.32, 43.00]	43.00	Source graph label	4.92	15.83
Son 2021	NC	T2	Buffer OR	(0.00, 12.43]	(12.43, 43.00]	43.00	Source graph label	6.06	18.63
Simoes 2022	Netherlands	T2	HR/count	(0.00, 0.31]	(0.31, 10.00]	10.00	Netherlands proxy region	0.20	0.70
Simoes 2022	Netherlands	T2	HR/count	(0.00, 0.62]	(0.62, 10.00]	10.00	Same study/source maximum	0.36	0.95
Simoes 2022	Netherlands	T2	HR/count	(0.00, 0.93]	(0.93, 10.00]	10.00	Same study/source maximum	0.47	1.23
Simoes 2022	Netherlands	T2	HR/count	(0.00, 1.24]	(1.24, 10.00]	10.00	Same study/source maximum	0.53	1.52

T1 denotes an explicit support pair and T2 a proxy support mapping. Role identifies restriction eligibility: Explicit OR and Point OR records enter both restrictions, Buffer OR records enter the OR-only restriction but not the explicit-support restriction, and HR/count records enter only the primary working analysis.

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